

MALABO, Equatorial Guinea

WMO: 648100

Lat: 3.755N

Lon: 8.709E

Elev: 23

StdP: 101.05

Time Zone: 1.00 (E01)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.8	21.9	19.8	14.5	24.1	20.1	14.9	24.0	7.6	27.0	7.2	26.9	1.5	230	0.312

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.3	32.2	26.6	32.0	26.6	31.6	26.5	28.2	30.2	27.8	30.0	27.5	29.9	4.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.9	24.1	29.6	27.2	23.0	29.1	27.0	22.7	29.0	91.3	30.4	89.0	30.9	87.2	30.1	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
6.7	6.0	5.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			DB	20.5	33.2	0.5	0.4	20.1	33.4	19.8	33.6	19.5	33.8	19.1	34.1	
			WB	19.2	29.4	0.9	1.1	18.6	30.2	18.1	30.9	17.6	31.5	16.9	32.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	27.2	27.8	27.7	27.3	27.1	26.3	25.6	25.4	25.6	26.0	26.5	26.9
	DBStd	1.14	0.83	0.92	0.94	0.93	0.83	0.81	0.78	0.64	0.74	0.72	0.77	0.69
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6066	533	499	550	519	531	488	485	478	467	496	495	525
	CDD18.3	3024	275	266	292	269	272	238	227	219	217	237	245	267
	CDH23.3	26967	2839	2903	3157	2776	2621	1894	1498	1324	1398	1764	2156	2637
CDH26.7	7897	1042	1082	1146	958	827	418	203	148	188	353	604	928	
Wind	WSAvg	2.4	2.4	2.8	2.5	2.2	2.0	2.1	2.4	2.9	2.6	2.3	2.3	2.2
Precipitation	PrecAvg	2537	45	74	160	214	276	286	306	291	325	332	176	51
	PrecMax	3037	139	256	253	322	394	501	463	470	556	478	386	119
	PrecMin	1858	4	5	46	113	183	180	81	54	194	243	51	6
	PrecStd	235	30	52	51	48	55	69	105	99	79	48	70	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	33.0	33.0	32.6	32.2	31.2	30.1	29.2	29.9	30.2	31.2	32.1
		MCWB	26.3	26.6	26.6	26.8	26.6	26.9	25.9	25.4	25.6	26.1	26.8	26.1
	2%	DB	31.9	32.2	32.2	32.0	31.4	30.2	29.1	28.8	29.1	29.9	30.9	31.2
		MCWB	26.3	26.6	26.7	26.8	26.7	26.3	25.5	25.2	25.5	26.0	26.6	26.1
	5%	DB	31.2	31.9	31.9	31.2	31.0	29.8	28.4	28.1	28.3	29.1	30.2	31.0
		MCWB	26.1	26.6	26.7	26.7	26.7	26.1	25.1	24.9	25.2	25.8	26.3	26.1
	10%	DB	30.9	31.1	31.1	30.9	30.2	29.0	27.9	27.6	27.9	28.8	29.8	30.6
		MCWB	26.0	26.5	26.7	26.7	26.5	25.8	24.9	24.7	25.1	25.7	26.2	26.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.0	28.7	28.6	28.5	28.6	28.0	26.9	27.0	27.0	27.9	28.5	28.2
		MCDB	30.3	31.1	31.1	30.5	30.3	29.7	28.5	27.8	27.9	28.9	29.7	30.0
	2%	WB	27.2	27.8	28.0	27.9	27.8	27.1	26.2	26.1	26.2	27.0	27.5	27.2
		MCDB	30.2	30.4	30.4	30.2	29.8	29.0	28.0	27.4	27.6	28.3	29.4	30.2
	5%	WB	26.8	27.2	27.4	27.5	27.3	26.6	25.8	25.5	25.9	26.3	26.9	26.8
		MCDB	30.0	30.4	30.3	30.1	29.7	28.5	27.6	27.1	27.5	28.2	29.3	30.0
	10%	WB	26.5	26.8	26.9	27.0	26.9	26.1	25.3	25.1	25.5	25.9	26.4	26.4
		MCDB	29.7	30.0	30.1	29.9	29.4	28.1	27.1	26.8	27.1	28.0	28.8	29.6
Mean Daily Temperature Range	5% DB	MDBR	7.9	7.3	7.0	7.0	6.8	5.9	5.0	4.8	5.0	5.8	6.6	7.8
		MCDBR	8.3	7.9	7.7	7.7	7.7	7.2	5.9	5.6	5.7	6.5	7.5	8.4
		MCWBR	4.5	4.1	3.9	4.3	4.3	4.2	3.4	3.2	3.4	4.0	4.4	4.7
	5% WB	MCDBR	7.8	7.5	7.2	7.3	7.0	6.5	5.5	5.2	5.4	6.0	7.0	7.9
		MCWBR	4.4	4.3	4.1	4.4	4.3	4.3	3.5	3.3	3.6	4.1	4.5	4.7
Clear-Sky Solar Irradiance	taub	0.804	0.820	0.788	0.559	0.500	0.532	0.537	0.533	0.455	0.460	0.577	0.734	
	taud	1.545	1.527	1.585	2.063	2.248	2.138	2.122	2.133	2.379	2.383	2.002	1.656	
	Ebn at Noon	597	603	626	771	798	760	762	782	861	862	756	634	
	Edh at Noon	288	299	282	171	138	151	155	158	126	125	181	254	
All-Sky Solar Radiation	RadAvg	4.85	4.62	4.42	4.49	4.17	3.35	3.03	3.03	3.32	3.79	4.17	4.63	
	RadStd	0.33	0.29	0.29	0.16	0.26	0.32	0.27	0.22	0.24	0.18	0.20	0.21	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A